

Group Lab: Case Study, Section 1

CAP3321C Data Wrangling | Miami Dade College | Professor Marquez

Today your group becomes an analytics team. A boss hands you a data file and five questions. There are no step-by-step instructions, no expected output to match, and more than one right answer for some questions. Your job is to load the data, decide how to clean it, answer the questions with tables and plots, and write what the results mean in plain English.

This is not graded. It is practice for your final, which is section 2 of the same case study. The notebook you build today is the starter file you will extend, so build it like something you want to come back to.

WHY ARE WE DOING THIS?

Every exercise so far told you which function to call. Real analysis starts with a question, not a function. Today you pick the function, and you also pick what "typical" means, whether a weird row stays or goes, and whether a table or a chart tells the story better. Getting comfortable with those choices is the difference between someone who knows pandas and someone who can do the job.

Pick your case study

Three datasets, three scenarios, five questions each. Same skills (chapters 2 through 4), different amounts of friction. Pick as a group.

Case study	Data	Difficulty	What makes it that difficulty
Laptops	<code>laptops.csv</code> , 205 rows	Easier	Small file, every question maps to one or two pandas calls. Two columns are text that look like numbers.
Medical Costs	<code>insurance.csv</code> , 1,338 rows	Medium	Clean file, but three questions ask "how is X related to Y" and you have to choose the right plot and read the pattern.
Airports	<code>tsa_claims1.csv</code> , 94,848 rows	Hardest	Big file, outliers that break averages, one question that needs two columns grouped together, and one where you have to define the metric before you can compute it.

COMMON MISTAKE

Picking Airports because it sounds interesting and then discovering at hour two that the group is still arguing about what counts as approved. If you take Airports, read its section below before you start and agree on your definitions early. If your group is unsure, Medical Costs is the best all-around choice.

Roles

DO THIS before anyone writes code

Assign roles. In a group of three, one person doubles up.

Lead Coder shares the screen and types. Everyone else contributes, but one person drives.

Data Interpreter owns the markdown. Writes the summary under every answer and the cleaning explanation. Asks "so what does this tell the boss?" after every result.

QA Reviewer reads the question out loud before each answer, checks that the answer actually answers it, and runs *Restart and run all* before the share-out.

Presenter chooses the one result the group will show the class and explains it in under three minutes.

Timeline

Time	What your group is doing
0:00 to 0:15	Form groups, pick a case study, assign roles, open the starter notebook, run the setup cell.
0:15 to 1:00	Explore and clean. <code>info()</code> , <code>describe()</code> , <code>value_counts()</code> . Decide what to fix and write down why.
1:00	Checkpoint. See the red box below.
1:00 to 2:15	Answer the five questions. Heading, code, summary, repeat.
2:15 to 2:35	Polish. Every summary written, every cleaning decision explained, notebook runs top to bottom.
2:35 to 2:50	Share-out. Each group presents one question.

STOP AND CHECK at the one-hour mark

Before you go further, your group should be able to show the instructor three things: the cleaned DataFrame with its shape, the markdown cell explaining what you cleaned and why, and the heading for Question 1. If you do not have all three, flag it now. An hour of exploring is plenty; an hour and a half is a warning sign.

What every notebook needs

From the Murach case study guidelines, adapted for Colab:

- A markdown heading for every question.
- A table or a plot that answers the question.
- A short written summary under each answer. This is the part your boss reads.
- Comments or markdown that explain any cleaning decision and any code that is not obvious.
- The notebook runs top to bottom with *Restart and run all* and produces no errors.

WHY ARE WE DOING THIS?

A notebook full of correct code and no words is a puzzle, not a report. Your final builds on this file, and in three weeks you will not remember why you dropped those rows. Write it down now, for the boss and for future you.

Laptops

Easier

Scenario. You work for a hardware company about to enter the laptop market. Your boss wants a report that answers five questions about the current market.

Starter notebook: `case-study-laptops-starter.ipynb`

What to look at before answering anything

- `ratings_5max` is text that looks like `3.3 / 5`. `disk_space` is text like `256 GB SSD` with a leading space. Neither is needed for section 1, so you can leave them alone, but note it: section 2 will need them.
- One row has a missing `laptop_name`. Decide whether it stays.
- There are two price columns. `discount_price` is what a buyer pays today; `old_price` is the list price. Pick one, say which, and use it everywhere.

The questions

1. How many different laptop brands are there?
2. What are the names and prices of the most and least expensive laptops?
3. How are laptop prices distributed?
4. What are the min, max, and mean display sizes?
5. What is the average price for each brand?

STOP AND CHECK

Setup shape is `(205, 9)`. Question 1 should come out to **9 brands**. If you get a different count, look at the unique values for stray spaces or capitalization.

COMMON MISTAKE

Answering Question 5 with a table sorted alphabetically. Sort by price instead. Your boss wants to see who is at the top of the market and who is at the bottom, not the alphabet.

Medical Costs Medium

Scenario. You work for a health care company. Your boss wants a report that answers five questions about medical insurance charges.

Starter notebook: `case-study-medical-starter.ipynb`

What to look at before answering anything

- This file is clean: no missing values, correct types. Check anyway with `info()` and `isna().sum()`, then write "we checked and found nothing to fix" in your cleaning cell. That is a real finding.
- Look at the unique values of `sex`, `smoker`, and `region` to confirm there are no typos.

The questions

1. How is age related to medical costs?
2. How is number of children related to medical costs?
3. How is the number of people distributed by region?
4. How is the number of people distributed by age?
5. How are the charges distributed?

STOP AND CHECK

Setup shape is `(1338, 7)`. Question 3 should show four regions with the southeast slightly ahead of the other three. If one region is missing, your cleaning removed rows it should not have.

COMMON MISTAKE

Plotting Question 1 as a line chart. Age and charges are two numeric columns with one dot per person, so that is a scatter plot. When you see it, you will notice the dots form separate bands. Say so in your summary and take a guess at what separates them. You will find out in section 2.

Airports Hardest

Scenario. You are an analyst at the TSA. Your boss wants a report that answers five questions about insurance claims filed against U.S. airports.

Starter notebook: `case-study-airports-starter.ipynb`

What to look at before answering anything

- Run `value_counts()` on `Status`, `Claim Type`, and `Claim Site`. Some categories have only a handful of rows out of nearly 95,000. Decide whether they belong in your analysis and write down why.
- Run `describe()` on `Claim Amount`. Look at the max, then look at the mean, then look at the

median. Those three numbers tell you why "typical" is a decision, not a calculation.

- The date columns loaded as text. You do not need them for section 1. Note it and move on; section 2 will ask you to convert them.

The questions

1. What is the most common type of insurance claim?
2. Which claim site within the airport are claims most commonly filed for?
3. What type of claim is made most at each claim site?
4. What is the typical claim amount?
5. What is the overall claim approval rate for the entire U.S.?

STOP AND CHECK

Setup shape is `(94848, 13)`. Question 1 should show **Passenger Property Loss** as the clear leader with roughly 60,000 claims. If your top category is something else, you loaded the wrong column.

COMMON MISTAKE

Reporting the mean claim amount as "typical." One claim in this file is for \$125 million. It drags the mean to over \$3,600 while the median is under \$200. Show both, explain the gap, and tell the boss which one to trust.

COMMON MISTAKE

Computing the approval rate without defining it. `Status` has Approved, Denied, and Settled as the big three. Is a settled claim a win for the passenger? Decide as a group, state the definition in markdown, then compute. A group that says "we counted Approved only and here is why" is doing better analysis than a group that quietly picks one.

WHY ARE WE DOING THIS?

Question 3 is the one that separates this study from the other two. "Most common type at each site" cannot be answered by one `value_counts()`. You need to group by site and then find the top type inside each group, or build a crosstab of site against type and read across the rows. Either way, you are combining two columns for the first time. That is the core skill of chapters 5 through 8, so you are getting a head start.

Share-out

Each group gets three minutes. The Presenter shows one table or plot and answers: what question does it answer, what does it say, and what surprised you? Pick the result your group argued about the most. Those make the best presentations.

Before you leave

DO THIS

1. Runtime menu, then *Restart and run all*. Fix anything that breaks.
2. Every group member downloads the notebook (File menu, then *Download .ipynb*). This is your starter file for the final.
3. Save it somewhere you will find it in three weeks.

Troubleshooting

What you see	Likely cause	Fix
<code>NameError: name 'pd' is not defined</code>	Setup cell has not run in this session	Run the setup cell again
<code>KeyError: 'Claim type'</code> (or any column name)	Column names are case-sensitive and some have spaces	Print <code>df.columns</code> and copy the name exactly
Histogram is one tall bar and a lot of empty space	An outlier is stretching the x-axis	Filter to a sensible range for the plot, and say in markdown that you did
<code>TypeError</code> when computing a mean	The column is text, not numbers	Check <code>info()</code> ; convert with <code>astype(float)</code> or <code>pd.to_numeric()</code> after cleaning the text
Notebook works cell by cell but fails on Restart and run all	A cell depends on something you ran earlier and later deleted	Read the error, find the missing variable, put its definition back above where it is used